Modeling Determinants of Population Health

Chapter 4
Eric Delmelle
October 2, 7 2025

Chapter Overview

- Understanding why some people are healthier than others is a core focus of public health.
- Health is shaped by multiple factors, including:
  - Genetics
  - Lifestyle choices
  - Environment
  - Social conditions
- This presentation explores models explaining how determinants interact to influence population health.

What are determinants of health?

- **Determinants of health** are factors that influence an individual's or a population's health.
- These determinants are often categorized into **five key groups**.
- Understanding these determinants helps in designing effective health policies and interventions.

Biological and Genetic Determinants

- **Genetics:** Inherited traits that influence susceptibility to diseases (e.g., family history of diabetes or heart disease).
- **Biological Factors:** Age, sex, and genetic conditions (e.g., Down syndrome, cystic fibrosis).
- **Example:** A person with a genetic predisposition to high cholesterol may develop heart disease if environmental and behavioral factors also contribute.

Behavioral Determinants

- Individual **lifestyle choices** that impact health.
- Key behaviors include:
  - **Diet and Nutrition:** Poor diet leads to obesity and related diseases.
  - **Physical Activity:** Lack of exercise increases risk for cardiovascular diseases.
  - **Substance Use:** Tobacco, alcohol, and drug use significantly affect health outcomes.
- **Example:** Smoking is a leading cause of lung cancer and heart disease.

Social and Economic Determinants

- **Education and Income:** Higher levels of education correlate with better health outcomes.
- **Employment and Working Conditions:** Stable employment provides income and health benefits.
- **Social Support Networks:** Strong community ties improve mental and physical health.
- **Example:** Individuals in low-income communities often have limited access to healthy foods and healthcare services.

Environmental Determinants

- **Physical Environment:** Pollution, climate change, and access to clean water affect health.
- **Built Environment:** Availability of sidewalks, parks, and grocery stores.
- **Housing Conditions:** Overcrowding and poor sanitation increase disease risks.
- **Example:** High air pollution in urban areas is linked to increased rates of asthma and respiratory diseases.

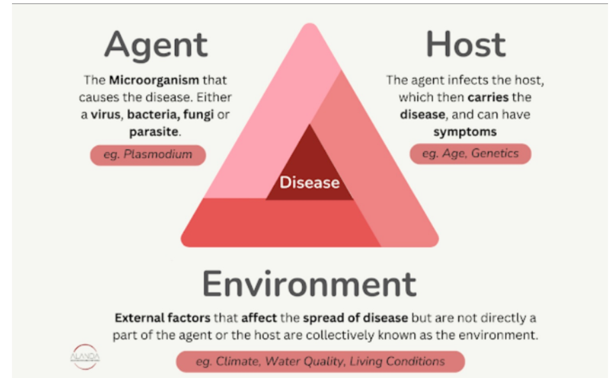
Healthcare Access and Quality

- **Availability of Healthcare Services:** Lack of access to preventive and emergency care worsens health disparities.
- **Health Insurance:** Affects affordability of medical care.
- **Cultural Competency:** Healthcare providers must address diverse patient needs.
- **Example:** Rural communities often experience healthcare shortages, leading to untreated conditions.

Why Do We Need Health Determinant Models?

- Health is not just about treating diseases—it's about **preventing** them.
- Important questions:
  - Why do wealthier neighborhoods have better health outcomes?
  - Why are chronic diseases more common in certain groups?
  - How do stress and social support affect mental health?
- Traditional **biomedical models** focus on treatment, but population health considers **broader social and environmental causes.**

1 The Epidemiologic Triad



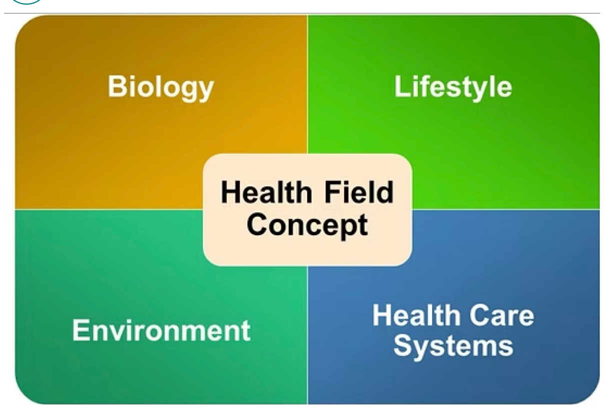
1 The Epidemiologic Triad

- A **basic model for infectious diseases** with three components:
  1. **Host** – Individual or population affected.
  2. **Agent** – Disease-causing factor (e.g., bacteria, viruses, smoking).
  3. **Environment** – External factors affecting exposure (e.g., air pollution, sanitation).
- **Example: Tuberculosis**
  - **Agent:** Mycobacterium tuberculosis
  - **Host:** Weakened immunity due to malnutrition
  - **Environment:** Overcrowding increases risk
- **Limitation:** Does not fully explain chronic diseases.

2 The Health Field Model

- Shifted focus from infectious diseases to **chronic conditions**.
- Identifies **four major health determinants**:
  1. **Human Biology** – Genetic and physiological factors.
  2. **Lifestyle** – Behaviors like smoking, diet, exercise.
  3. **Environment** – Social and physical surroundings.
  4. **Healthcare System** – Access and quality of medical care.

2 The Health Field Model

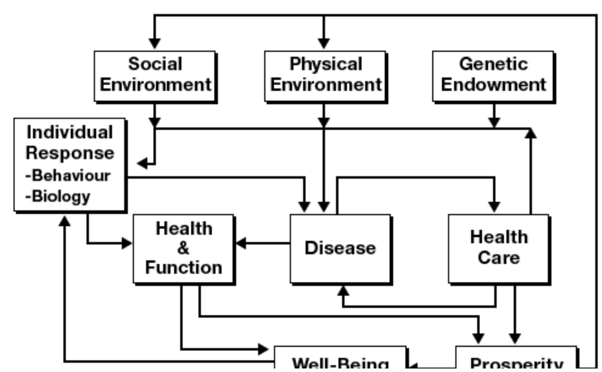


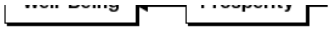
Example: Heart Disease

- **Lifestyle:** Smoking, unhealthy diet.
- **Environment:** Food deserts, limited access to healthy food.
- **Healthcare System:** Focuses on treatment rather than prevention.
- **Biology**: Family history of heart disease, genetic predisposition to high cholesterol

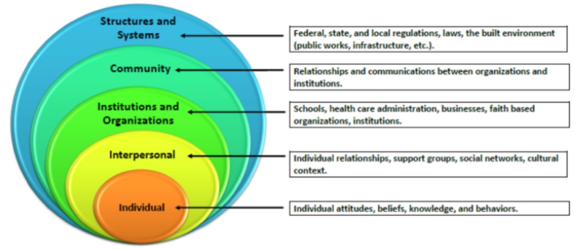
2 The Health Field Model

- A more **advanced** model





3 The Socio-ecological model



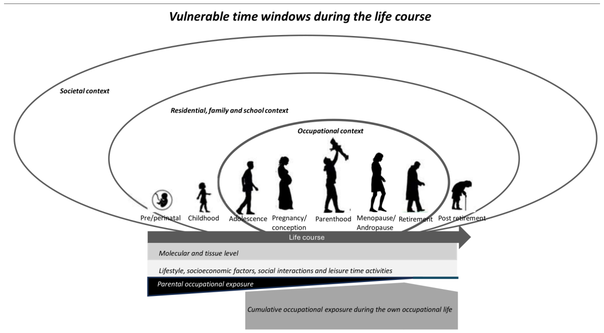
3 The Socio-ecological model

- **Health is shaped at multiple levels:**
  - **Individual** – Genetics, behavior.
  - **Interpersonal** – Family, social support.
  - **Community** – Schools, workplaces, healthcare access.
  - **Societal** – Laws, policies, economy.

Example: Obesity Prevention

- **Individual:** Encouraging healthy diet.
- **Community:** More grocery stores and parks.
- **Societal:** Taxes on unhealthy foods.

4 The Life Course Perspective



4 The Life Course Perspective

- Health is **cumulative** and shaped over a lifetime
- Early-life exposures impact later health outcomes
- Critical periods: Childhood, adolescence, pregnancy
- Protective and risk factors evolve through different life stages

4 The Life Course Perspective

- Examines how **early-life experiences shape future health**.
- **Key Concepts:**
  1. **Latency Effects** – Early-life exposures impact adult health.
  2. **Cumulative Effects** – Stress builds up over time.
  3. **Pathway Effects** – Disadvantages in childhood limit future opportunities.

Example: Childhood Poverty & Health

- Poor nutrition in childhood → Increased risk for diabetes, heart disease in adulthood.
- Chronic stress affects **brain development**.

4 The Life Course Perspective

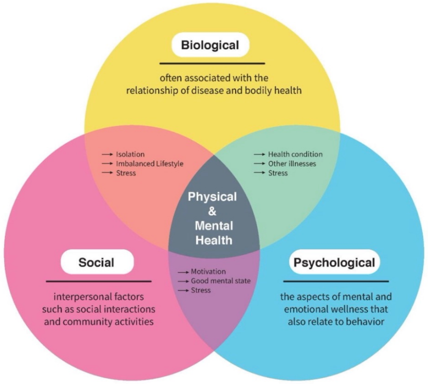
4 The Life Course Perspective

- **Can geographic information keep you healthy?** youtube-link
- *Bill Davenhall* from ESRI

5 Biopsychosocial Model

- Health is influenced by **biological, psychological, and social factors**
- **Biological:** Genetics, immune function, neurobiology
- **Psychological:** Stress, behavior, cognition
- **Social:** Environment, socioeconomic status, relationships
- **Interaction:** These domains interact dynamically, shaping health outcomes

5 Biopsychosocial Model



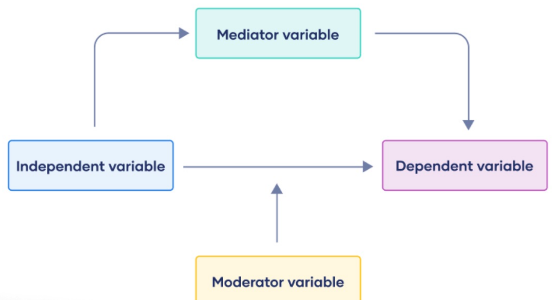
Pathways from Determinants to Health Outcomes

- **Upstream Determinants** (e.g., policy, environment) →
- **Intermediate Factors** (e.g., behaviors, stress) →
- **Downstream Outcomes** (e.g., disease, mortality)
- Example: **Socioeconomic Status → Diet & Stress → Cardiovascular Health**

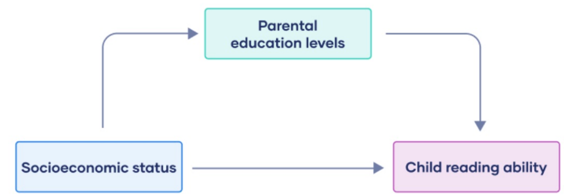
Pathway Analysis

- How do **various determinants interact** to influence health outcomes?
- **Mediators:** Factors that explain part of the relationship (e.g., stress mediates poverty → health)
- **Moderators:** Factors that modify the strength of the relationship (e.g., social support buffers stress effects)
- **Example:** Education → Health Literacy → Better Health Management

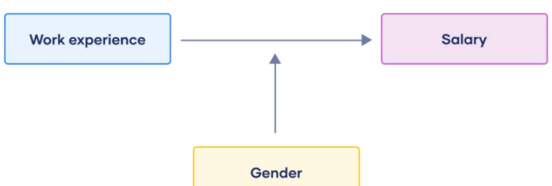
Pathway Analysis



Pathway Analysis



Pathway Analysis



Nonlinear Dynamics & Health

- Health systems are **complex and nonlinear**
- **Feedback Loops:**
  - Positive: Obesity → Inflammation → More Obesity
  - Negative: Stress → Social Support → Stress Reduction
- **Threshold Effects:** Small changes can lead to sudden shifts
  - Example: Chronic stress accumulation suddenly leading to burnout or illness

Policy Implications & Future Directions

- **Health policies should be evidence-based and proactive**.
- **Types of interventions:**
  1. **Upstream:** Tackling root causes (e.g., poverty reduction).
  2. **Midstream:** Changing risk behaviors (e.g., smoking bans).
  3. **Downstream:** Treating diseases after they occur.

Future trends

- **Precision Public Health** – Using **big data & genetics** for targeted interventions.
- **Climate Change & Health** – Understanding **environmental risks**.
- **Health Equity** – Ensuring **all populations** have access to resources.

Conclusions

- **Health is shaped by multiple determinants**, not just healthcare.
- **Interventions must be multi-level** to be effective.
- **Understanding health models** helps in designing **better policies**.

Reflection Questions:

- 1. What social determinants impact your own health?
- 2. How can policies create healthier communities?

Thank you!